

Unit 3

* Servlet Program to say "Hello World".

```
import java.io.*;
import java.servlet;
import javax.servlet.http;
public class FirstServlet extends HttpServlet
{
    public void doGet (Http Servlet Request request,
    Http Servlet Response response) throws IOException, Servlet
    Exception;
    {
        response. set Content Type ("text /html");
        Print Writer out = response. getWriter ();
        out. print In (" <html> ");
        out. print In (" <head> ");
        out. print In (" <title> My First Servlet </title> ");
        out. print In (" <body> ");
        out. print In (" <h1> Hello World </h1> ");
        out. print In (" </body> ");
        out. print In (" </html> ");
    }
}
```

* Session Management, Different techniques.

- Java Servlet are programs that run on a web or Application server & acts as a middle layer between a requests coming from a web browser or HTTP client and databases or applications on the HTTP server. It is a thread based technologies.

Session Management Technique

It is a way to maintain state of an user. HTTP protocol is a stateless so we need to maintain state using session management technique. It is used to recognize the particular user.

Techniques are

① Cookies

They are some little information that can be left on your computer by the other computer when we access an internet.

Using the information stored in the cookies, some advertising agencies can keep track of your internet usage, linking of users.

Syntax:

Cookie (string name, string value)

Types :

- i] Non - Persistent
- ii] Persistent Cookie

② Using Hidden form field

Hidden form fields are the input fields which are not displayed on the page but its value is sent to servlet as other input fields.

③ URL Rewriting

If the client has disabled cookies in the browser then session management using cookie won't work.

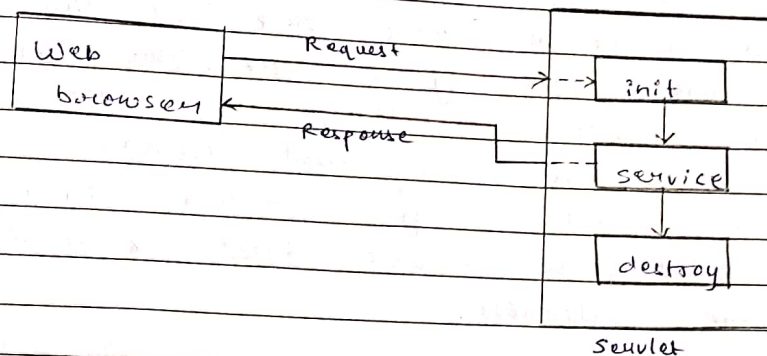
A token (parameter) is added at the end of URL, the token consist of name/value pair. separated by an equal (=) sign.

④ HTTP Session

HTTP session object is used to store entire session with a specific client.

We can store, retrieve & remove attribute from HttpSession object on clients first request, the web container generates a unique session ID & gives it back to the client with response.

* Servlet Life cycle



i. init () method

The server basically invokes the `init ()` method of servlet. This method is called only when the servlet is loaded in the memory for the first time. Using this method initialization parameters can also be passed to the servlet in order to configure itself.

ii. service () method

It is the main method to perform the actual task. The servlet container calls the `service ()` method to handle requests coming from the client & to write the response back to the client. When server receives a request, the `service ()` method checks the HTTP request-type & calls `doGet`, `doPost` methods as appropriate.

iii. destroy() method

It is called only once at the end of the life cycle of a servlet. This method gives your servlet a chance to close database connections, halt background threads & perform other such cleanup activities. After this method is called, the server object is marked for garbage collection.

* DTD with examples

1. DTD stands for Document Type Definition.
2. A DTD defines the structure & the legal elements and attributes of an XML document.
3. With a DTD, independent groups of people can agree on a standard DTD for interchanging data.
4. An application can use a DTD to verify that XML data is valid.

Types of DTD are:

- 1] Internal
- 2] External

Ex. -

If we want to put some information about some students in XML file then generally we use tag students followed by his/her name, address, standard and marks.

That means we are actually specifying the manner by which the information should be arranged in the XML file. And for this purpose the document type definition is used.

* style in XML

In XML, you can apply style using CSS or XSLT. XML itself is a data representation language and does not include style information natively, but you can link it to style instructions with CSS or XSLT.

XML Document

[book.xml]

⚡ ?xml -stylesheet type="text/css" href="style.css" ?

<library>

<book>

<title> Introduction to ML </title>

<author> John Smith </author>

<year> 2020 </year>

</book>

</library>

css (style.css)

title {

font-family : Arial, sans-serif;

color : blue;

font-weight : bold;

}

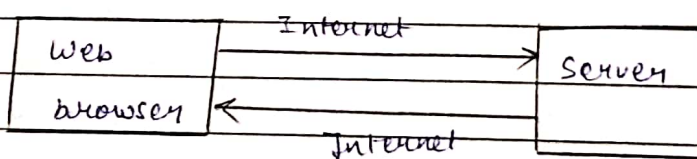
author {
font-family : Georgia, serif ;
color : green ;

}
year {
font-family : courier New, monospace ;
color : red ;

}

→ AJAX

1. AJAX is a Asynchronous JavaScript and XML.
2. Here,
 - Asynchronous means the execution of script does not disturb the user's work.
 - Javascript because, it makes use of JavaScripting to do the actual work.
 - XML because along with JS the XML is also supported to perform the given task in AJAX.
3. It is not a programming language.



4. When user makes a request, the browser creates an object for the XMLHttpRequest and a request is made to server over internet.

5. The server processes this request and sends the requested data to the browser.
6. At the browser side the returned data is processed using Javascript and the web document gets updated accordingly by sending the appropriate response.